

Temperature-dependent nanoscale conformations of mechanosensitive protein BmrA

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Keywords:

Membrane curvature; ABC Transporter; smFRET; ATP-ase activity; lipid membrane Mechanics

ABSTRACT

Membrane proteins are essential for exchanges across the impermeable lipid bilayer and operate by switching between distinct conformational states in response to stimuli such as ligand binding, ATP hydrolysis, or changes in the physical properties of the membrane. Their conformational equilibria are therefore highly sensitive to temperature and membrane mechanics.

We focus on BmrA, a bacterial ABC transporter that alternates between inward and outward-facing conformations during its ATP hydrolysis cycle. Biochemical ensemble measurements revealed that the ATPase activity of BmrA increases exponentially with temperature, suggesting a temperature-dependent shift in its conformational equilibrium.

Using single-molecule fluorescence resonance energy transfer (FRET) on membrane reconstituted BmrA proteins, we demonstrate the mechanosensitivity and temperature dependent remodeling of the conformational landscape of BmrA. We extract the transition rates between the major functional states and determine how thermal energy modulates the kinetics of conformational switching. This approach provides fundamental insight into the biophysics of membrane protein dynamics and clarifies the role of temperature and membrane mechanics in regulating the functions of transporter proteins.

REFERENCES

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